Data Analytics Project: Car Insurance Risk Prediction

Presented by

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This project focuses on predicting car insurance risk and characterizing car segments using a comprehensive dataset.

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Understanding Car Insurance Risk Data

Automobile Specifications

The dataset includes various characteristics of automobiles, such as make, fuel type, body style, engine specifications, and dimensions.

Risk Rating (Symboling)

Cars are assigned an insurance risk rating, or "symboling," which indicates how risky they are compared to their price. A value of +3 means risky, while -3 means safe.

Normalized Losses

This factor represents the average loss payment per insured vehicle year, normalized across specific size classifications.

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Key Project Objectives

Predict Risk Rating

Develop models to accurately predict the insurance risk rating (Symboling) of a car.

Characterize Car Segments

Identify and describe distinct segments of cars based on their risk profiles and inherent characteristics.

Deliverables

The project includes data cleaning, exploratory data analysis, predictive model training, dimensionality reduction, and car segmentation.

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Data Cleaning and Outlier Handling

Data Quality Evaluation

We evaluated data quality and checked for inconsistencies to ensure reliable analysis.

Outlier Mitigation

Potential outliers in features like Price, Horsepower, and Engine-size were observed.

IQR Capping Method

IQR Capping was applied to replace extreme values, reducing skewness without discarding valuable data.

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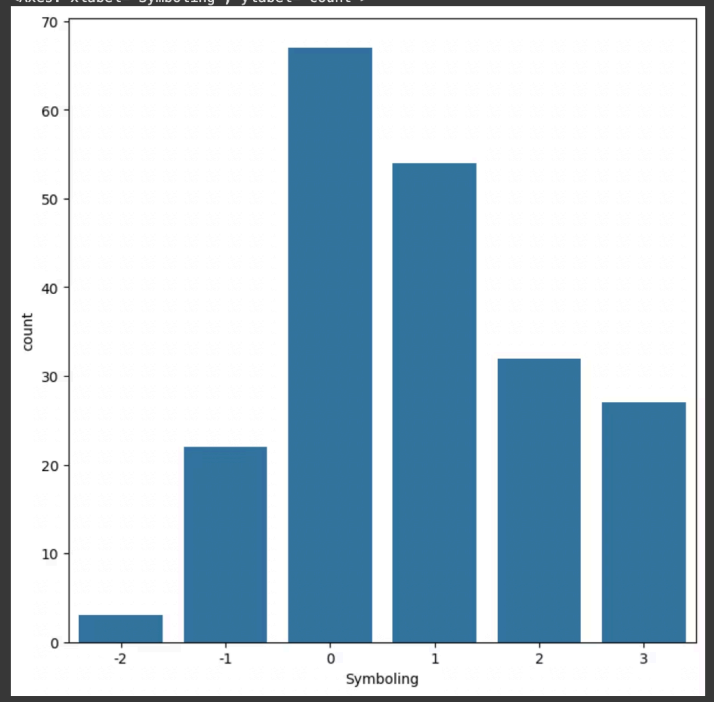
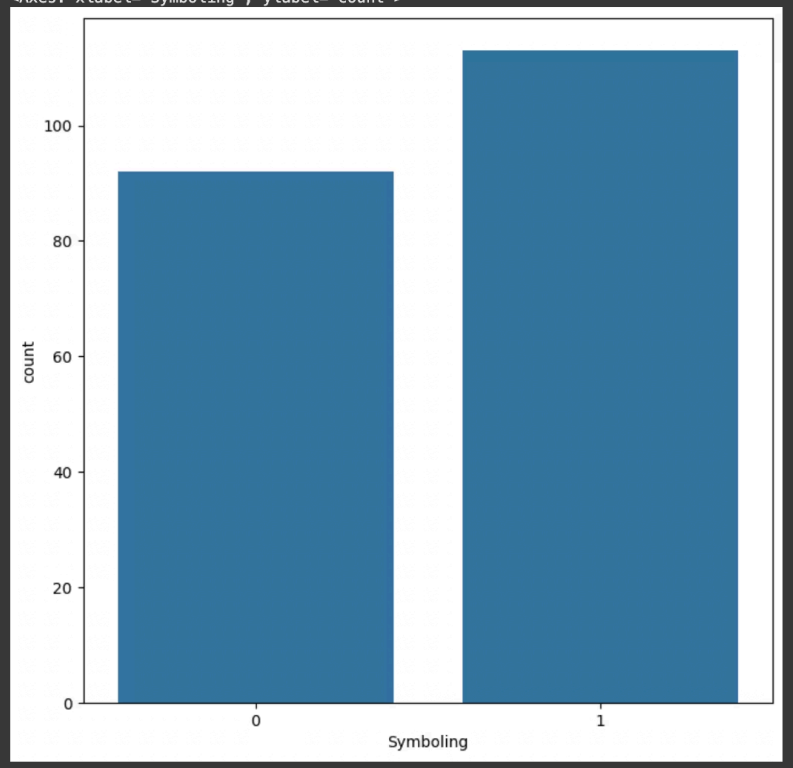
Symboling distribution

```
<Axes: xlabel='Symboling', ylabel='count'>
```



The dataset is quite unbalanced, so we should separate it between those that are above 0 and those that are equal to or less than 0 in the Symboling category.

```
<Axes: xlabel='Symboling', ylabel='count'>
```



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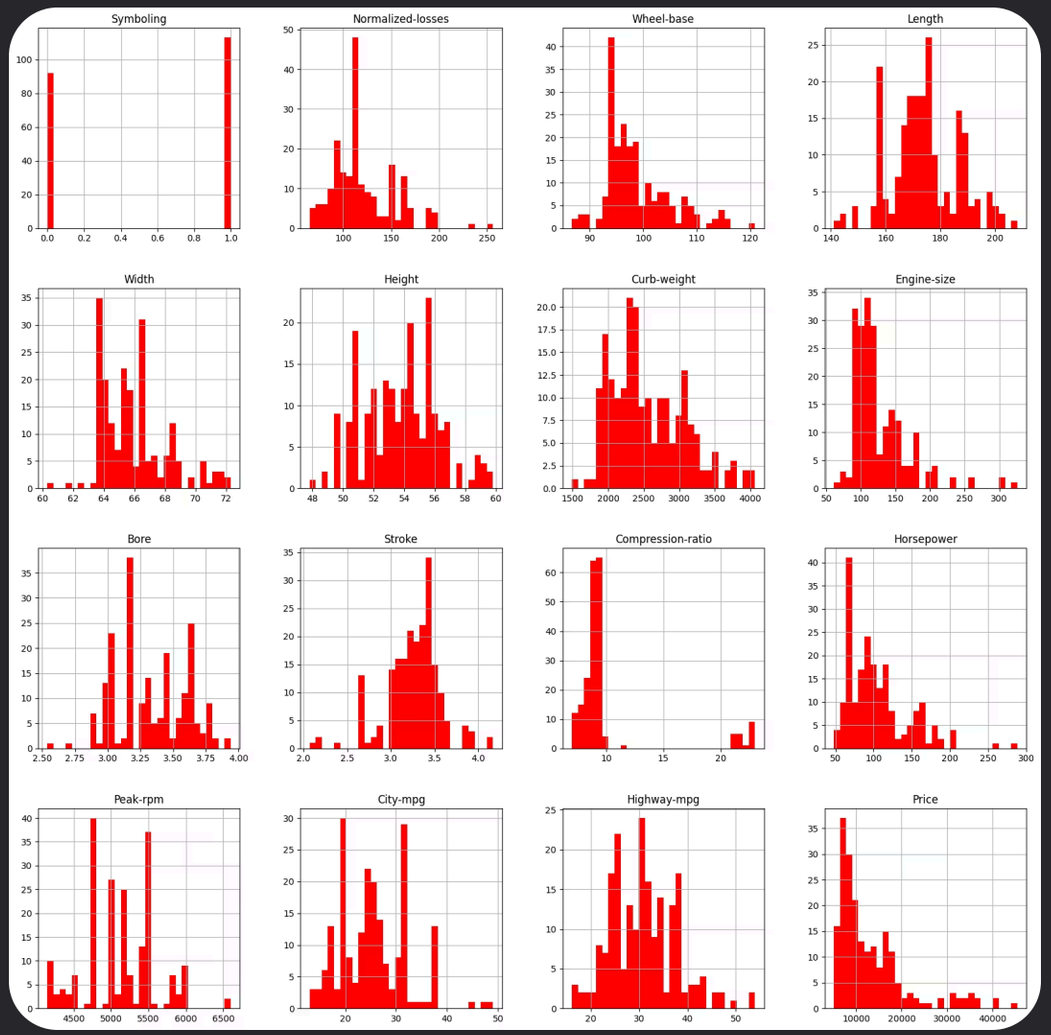
DISTRIBUTION ON EACH VARIABLE



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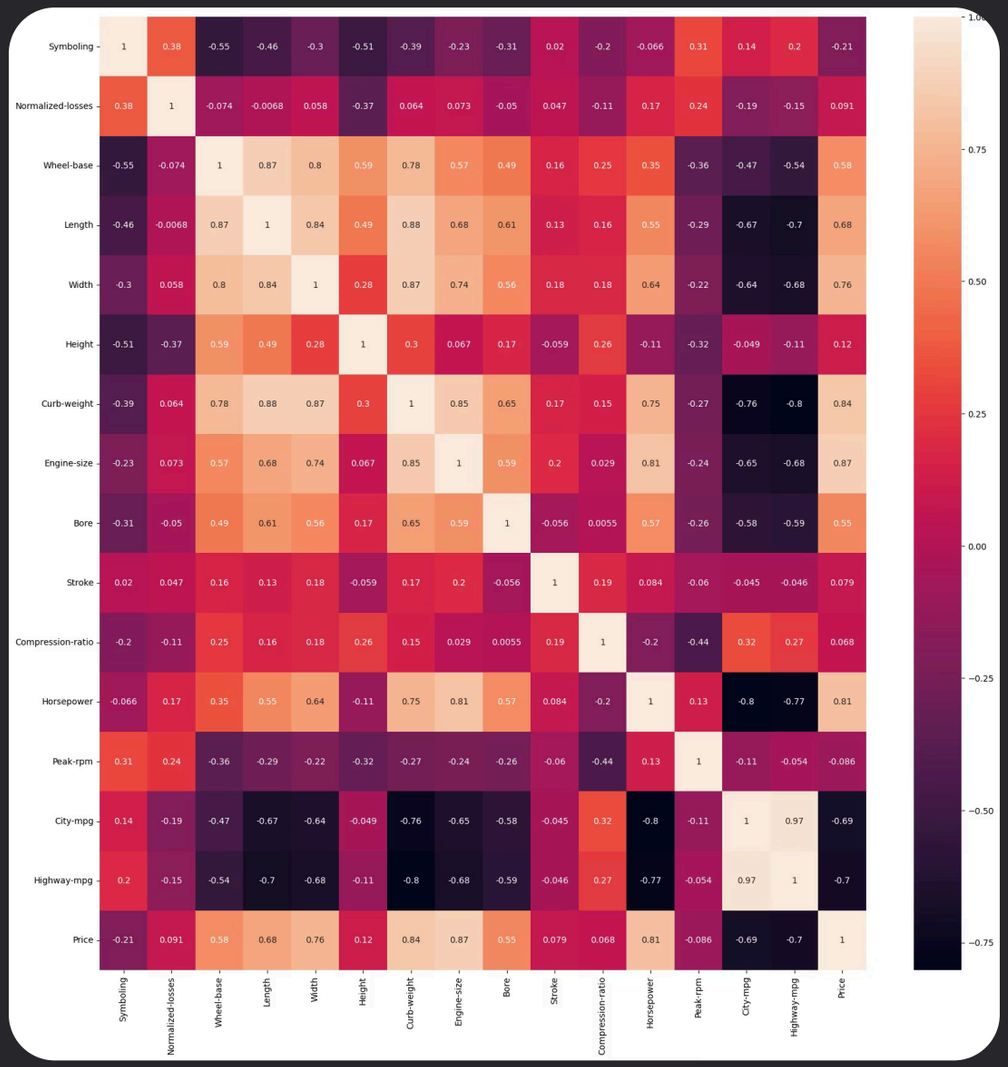
The correlation heatmap of each variable

The correlation heatmap highlights strong positive correlations between engine-size, curb-weight, and price, suggesting that larger and heavier vehicles tend to be more expensive. Meanwhile, city-mpg and highway-mpg show negative correlations with price, indicating that more fuel-efficient cars tend to be less expensive. These insights help identify key predictors for insurance risk or vehicle classification.



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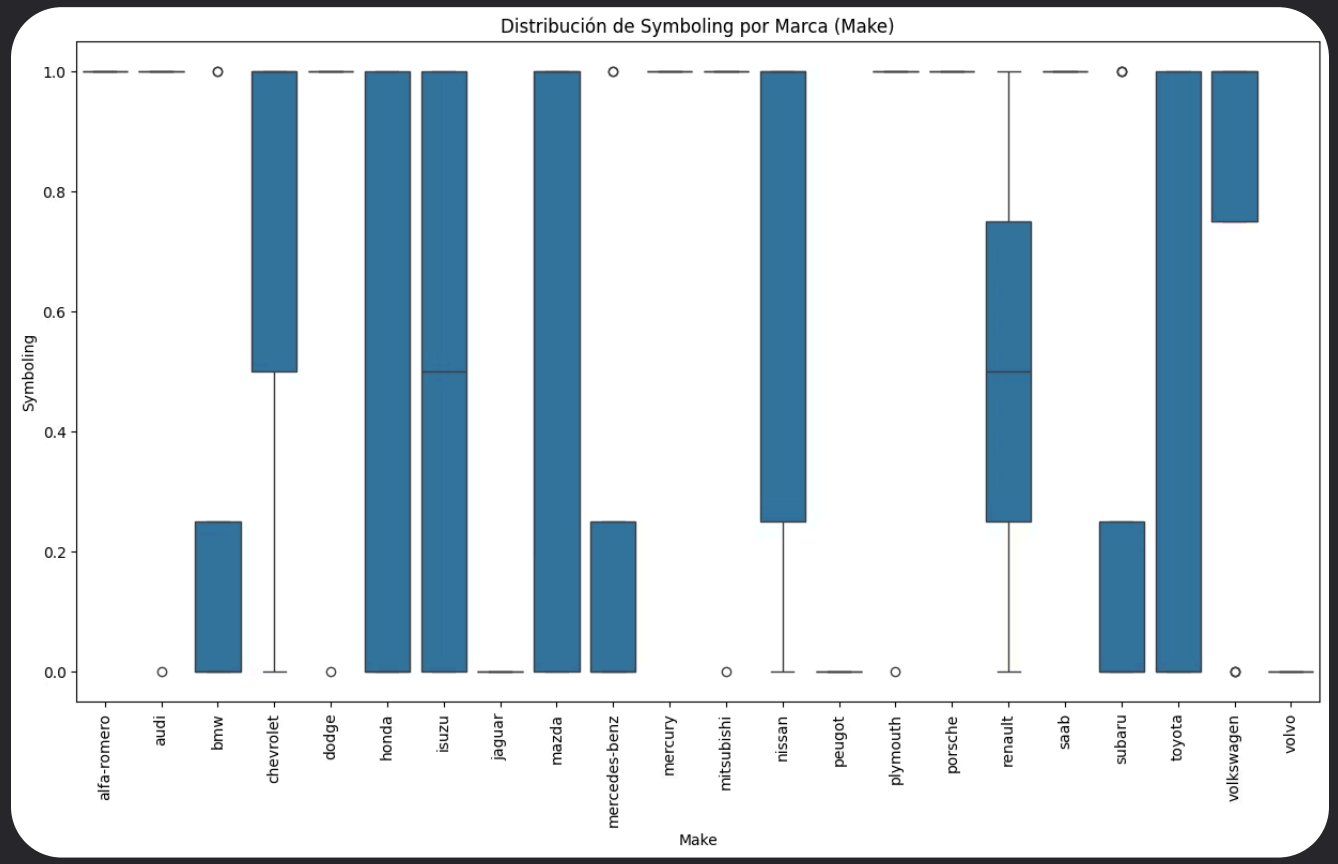
DATA VISUALIZATION



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Delivery #2

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Predictive Modeling for Risk

Data Preparation

Categorical variables were encoded, numerical variables scaled, and data split for training and testing.

Evaluation Metrics

Model performance was measured using metrics suitable for binary classification, focusing on accuracy and ROC AUC.

Model Training

Logistic Regression, Decision Tree, and Random Forest Classifiers were implemented and optimized.

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Selected Model

Based on the comprehensive evaluation across multiple relevant metrics, the **Tuned Random Forest Classifier** is selected as the best model for this task.

- **Highest ROC AUC:** Demonstrates the best overall discriminatory power between classes.
- **Low False Negatives:** Achieved the lowest reliable count (4) of the most critical error type (misclassifying risky cars as safe).
- **Excellent Balance (F1-Score):** Showed the best F1-score (0.89) among the reliably tuned models, indicating a strong balance between correctly identifying risky cars (Recall) and ensuring those predictions are accurate (Precision).
- **Robustness:** As an ensemble method, it is generally more stable and less prone to overfitting than a single Decision Tree, providing more confidence in its generalization ability. Although the baseline Decision Tree performed surprisingly well on this specific test set, its inherent instability makes the Random Forest a more reliable choice for deployment.

Performance Summary Table

Model	Accuracy	ROC AUC	F1-Score (Class 1)	Precision (Class 1)	Recall (Class 1)	False Negatives (FN)
Logistic Regression (Tuned)	0.7885	0.8711	0.81	0.82	0.79	6
Decision Tree (Baseline)	0.9038	0.9048	0.91	0.93	0.90	3
Decision Tree (Tuned)	0.7692	0.8396	0.76	0.90	0.66	10
Random Forest (Tuned)	**0.8846**	**0.9498**	**0.89**	**0.93**	**0.86**	**4**

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Deliverable #3

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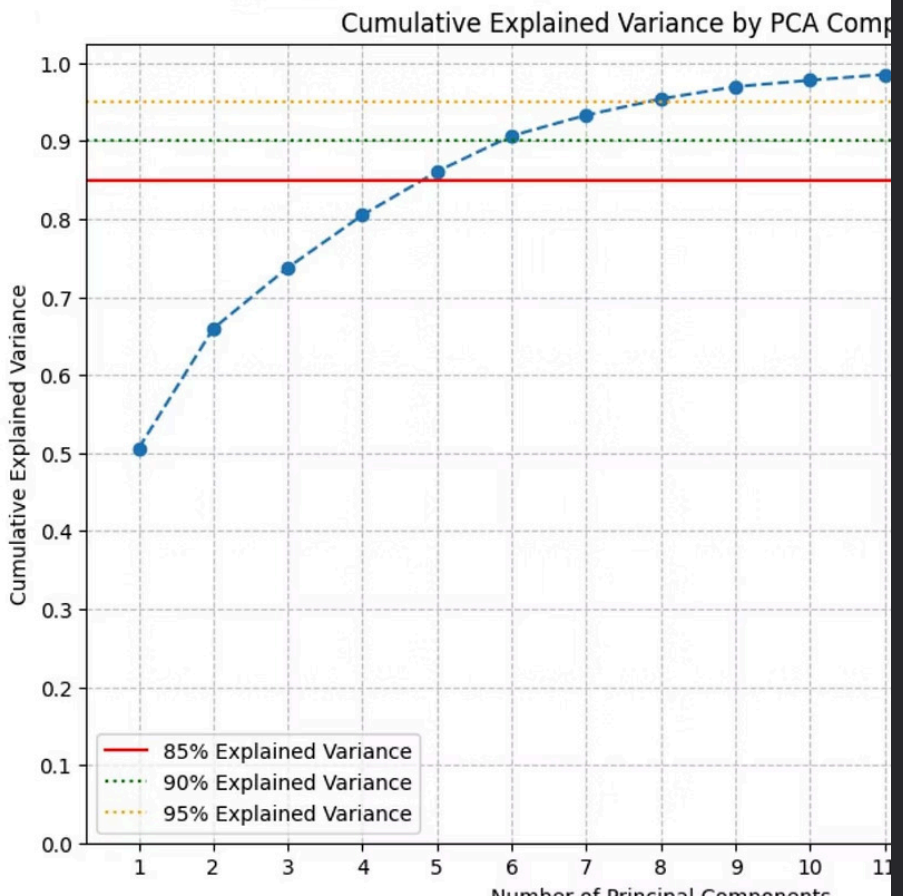
```
--- 2.1 Applying PCA and Visualizing Explained Variance -

Explained variance by each component:
  PC1: 0.5057 (Cumulative: 0.5057)
  PC2: 0.1537 (Cumulative: 0.6594)
  PC3: 0.0781 (Cumulative: 0.7375)
  PC4: 0.0675 (Cumulative: 0.8050)
  PC5: 0.0558 (Cumulative: 0.8608)
  PC6: 0.0457 (Cumulative: 0.9065)
  PC7: 0.0266 (Cumulative: 0.9331)
  PC8: 0.0207 (Cumulative: 0.9538)
  PC9: 0.0157 (Cumulative: 0.9696)
  PC10: 0.0082 (Cumulative: 0.9777)
  PC11: 0.0074 (Cumulative: 0.9851)
  PC12: 0.0053 (Cumulative: 0.9905)
  PC13: 0.0043 (Cumulative: 0.9947)
  PC14: 0.0036 (Cumulative: 0.9984)
  PC15: 0.0016 (Cumulative: 1.0000)
```



Dimensionality Reduction with PCA

PCA Application

Principal Component Analysis (PCA) was applied to reduce the number of numerical features.

Variance Retention

We aimed to preserve at least 85% of the original dataset's variance to simplify data and reduce noise.

Scaled Data

Appropriately scaled numerical features were used for PCA to ensure equal contribution from all features.

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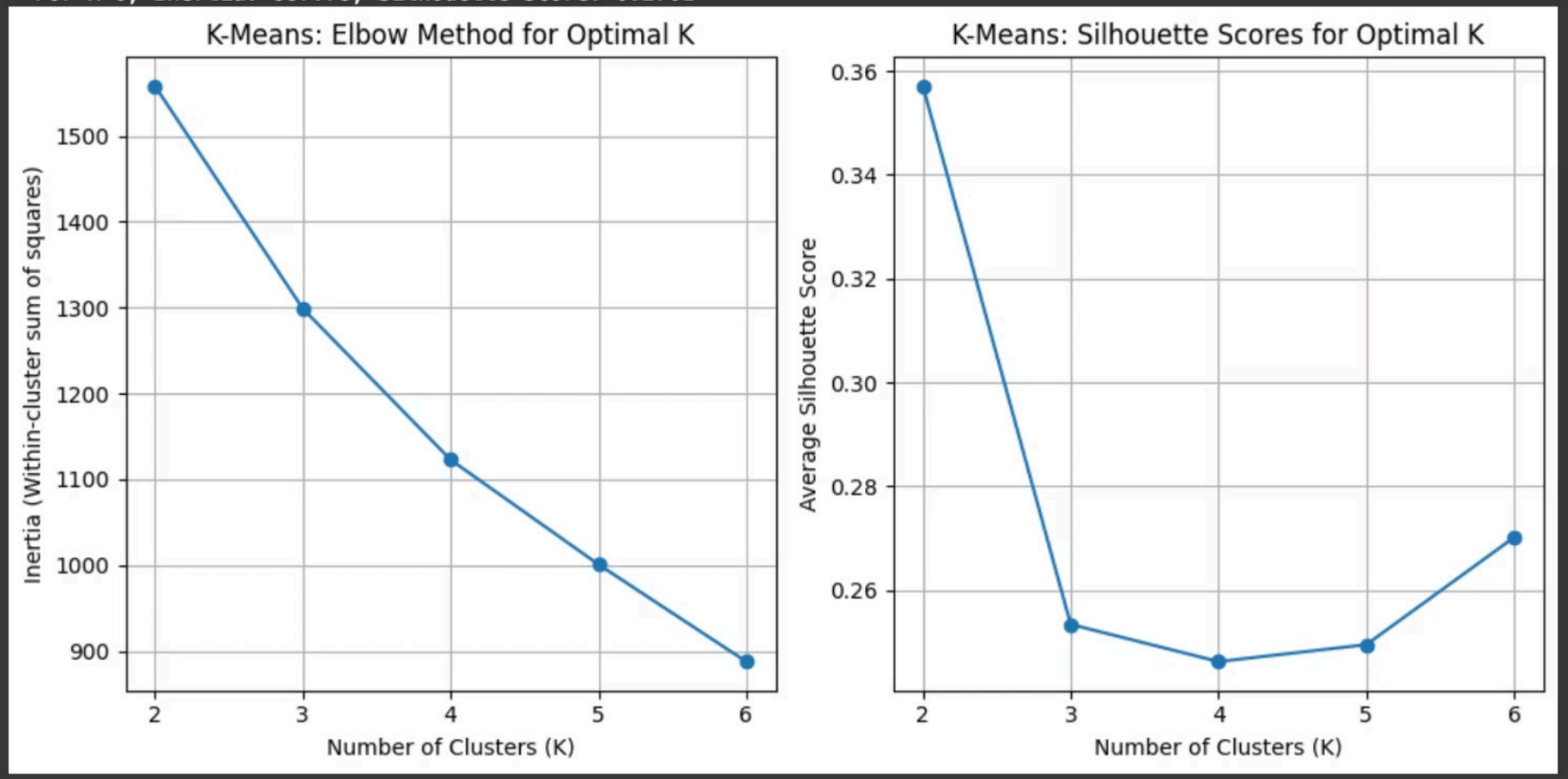
Finding the optimal K

```
--- 3.1 K-Means Clustering ---

Calculating Inertia and Silhouette Scores for K-Means...
  For K=2, Inertia: 1558.26, Silhouette Score: 0.3571
  For K=3, Inertia: 1299.04, Silhouette Score: 0.2534
  For K=4, Inertia: 1122.90, Silhouette Score: 0.2462
  For K=5, Inertia: 1001.00, Silhouette Score: 0.2495
  For K=6, Inertia: 887.78, Silhouette Score: 0.2701
```



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Car Segmentation with Clustering

K-Means Clustering

Used to partition data into distinct, non-overlapping groups. Optimal K was determined using Elbow and Silhouette methods.

Hierarchical Clustering

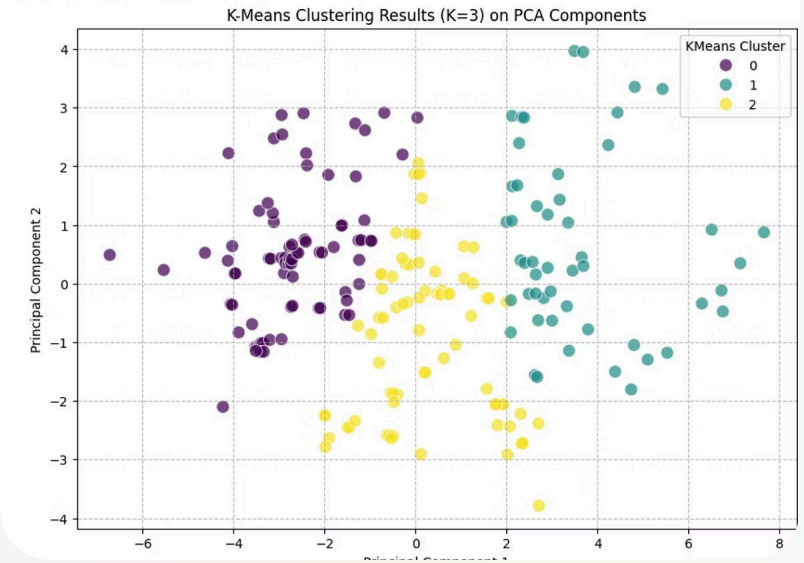
An alternative approach that builds a hierarchy of clusters, visualized with a dendrogram, allowing flexible cluster selection.

Segment Characterization

Clusters were analyzed based on original features (mean/median for numerical, mode/frequency for categorical) to understand unique characteristics.

```
Selected K for K-Means based on evaluation: 3

Number of cars in each K-Means cluster (K=3):
KMeans_Cluster
0    77
1    51
2    77
Name: count, dtype: int64
```



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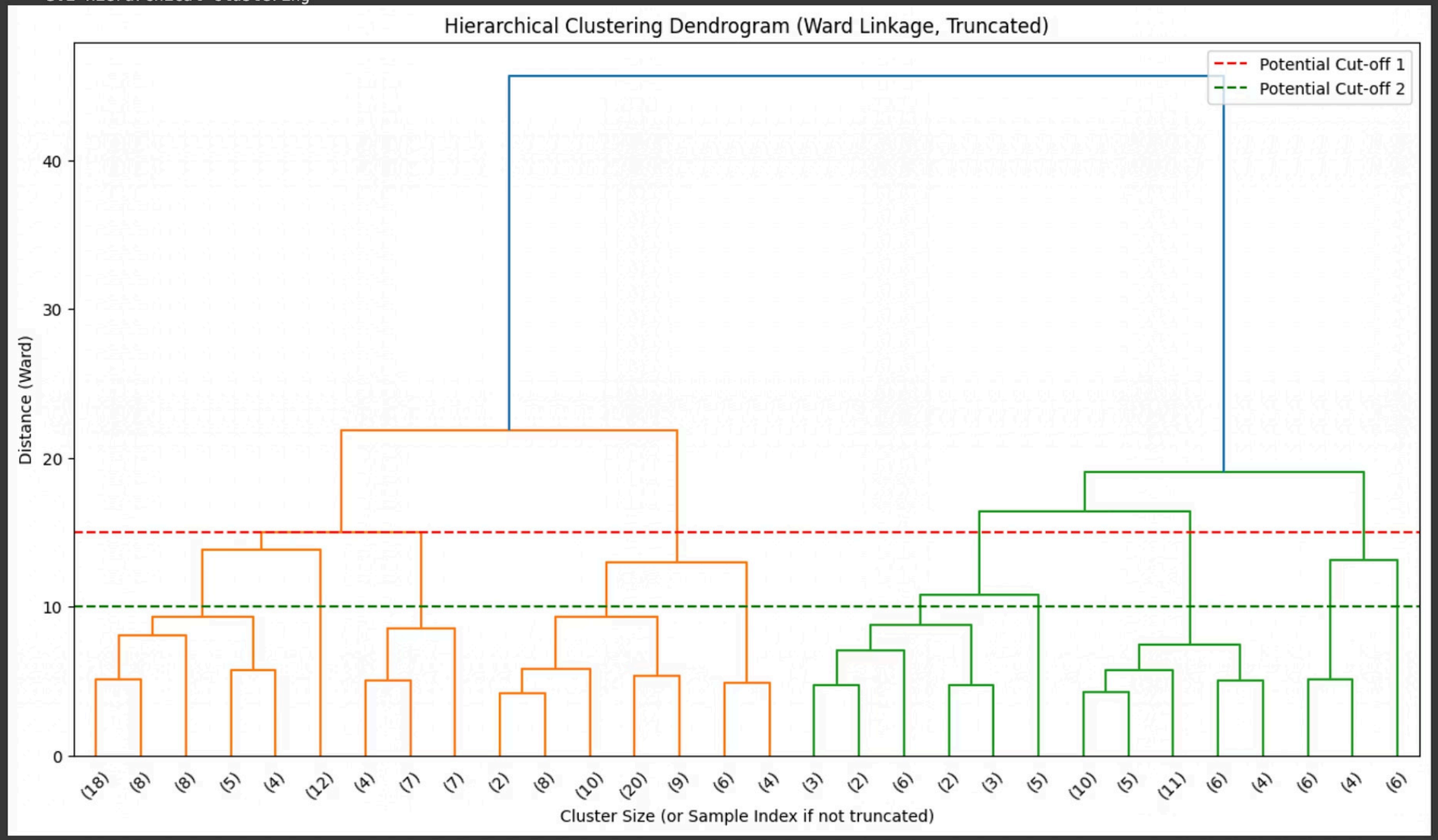
Dendogram

```
--- 3.2 Hierarchical Clustering ---
```



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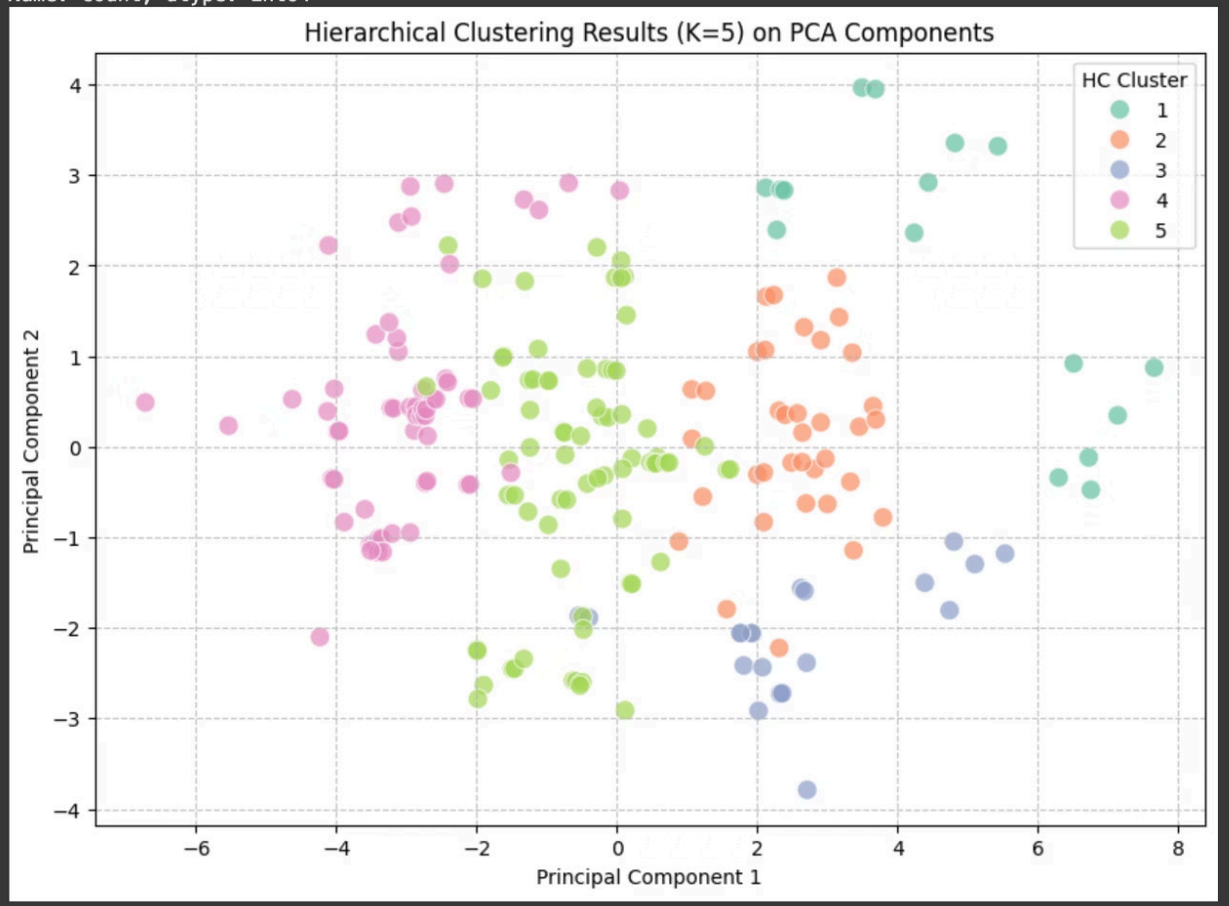
Clustering results

```
Selected K for Hierarchical Clustering based on dendrogram: 5

Number of cars in each Hierarchical Cluster (K=5):
1    16
2    36
3    21
4    59
5    73
Name: count, dtype: int64
```



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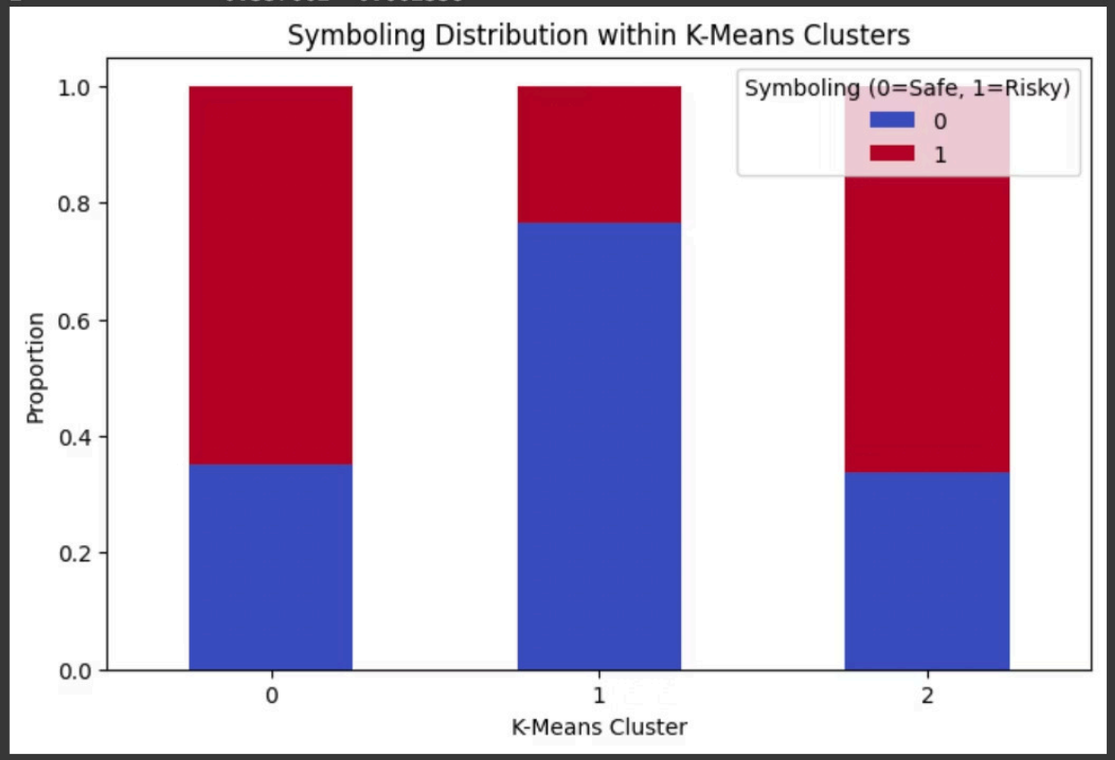
Segmentation

```
Distribution of original 'Symboling' within K-Means Clusters (Normalized by Cluster):
Symboling              0         1
KMeans_Cluster
0               0.350649  0.649351
1               0.764706  0.235294
2               0.337662  0.662338
```



Interpretation of K-Means Segment Characterization (K=3)

The K-Means clustering algorithm has grouped the dataset into three distinct segments. We will analyze each cluster based on its average numerical features, dominant categorical features, and risk profile (Symboling).

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Conclusion

The application of **Principal Component Analysis (PCA)** proved effective in reducing the dimensionality of the numerical features from 15 down to 5 principal components. These components managed to retain approximately 86% of the original dataset's variance, thereby simplifying the data structure for subsequent analysis without significant loss of information. This created a more manageable and potentially less noisy feature space for clustering.

Then, **clustering analysis** was performed using K-Means and Hierarchical Clustering techniques on the PCA-transformed data.

- For **K-Means clustering**, an optimal number of K=3 clusters was determined through an evaluation of the Elbow method and Silhouette scores.
- For **Hierarchical Clustering**, interpretation of the dendrogram (specifically a cut-off at a distance of 15) suggested an optimal K=5 clusters.

The primary focus of the **segment characterization** was on the three segments derived from K-Means. This analysis revealed distinct and meaningful car profiles:

- **Cluster 0 ("Economical, Compact, Higher-Risk Vehicles"):** Typically smaller, more fuel-efficient, and lower-priced cars (often Toyota sedans), which paradoxically showed a higher propensity for risk.
- **Cluster 1 ("Large, Powerful, Expensive, Safer Vehicles"):** Characterized by larger dimensions, more powerful engines, higher price points (often Peugeot sedans), and a generally lower risk profile.
- **Cluster 2 ("Mid-Range, Higher-Risk Hatchbacks"):** Representing vehicles with mid-range characteristics (often Toyota hatchbacks), generally more expensive than Cluster 0 but also exhibiting a higher risk profile.

In conclusion, we aplied unsupervised learning techniques to discover patterns and structures within the car dataset. By reducing dimensionality and then segmenting the vehicles, we identified distinct groups based on their inherent characteristics, offering valuable insights into different market segments and their associated risk profiles without reliance on pre-defined labels. This provides a strong foundation for understanding customer types and informing potential business strategies.

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